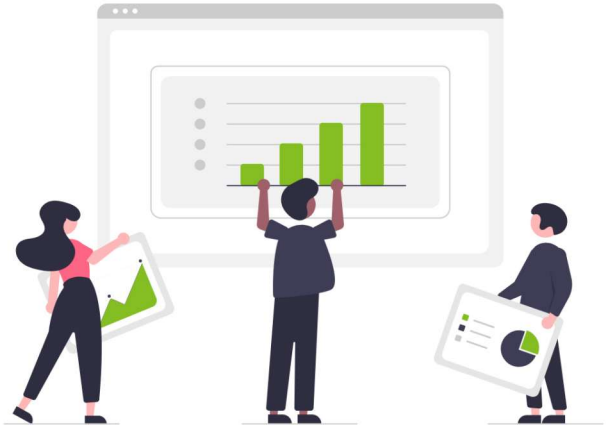

Task Guide:
Data Analysis

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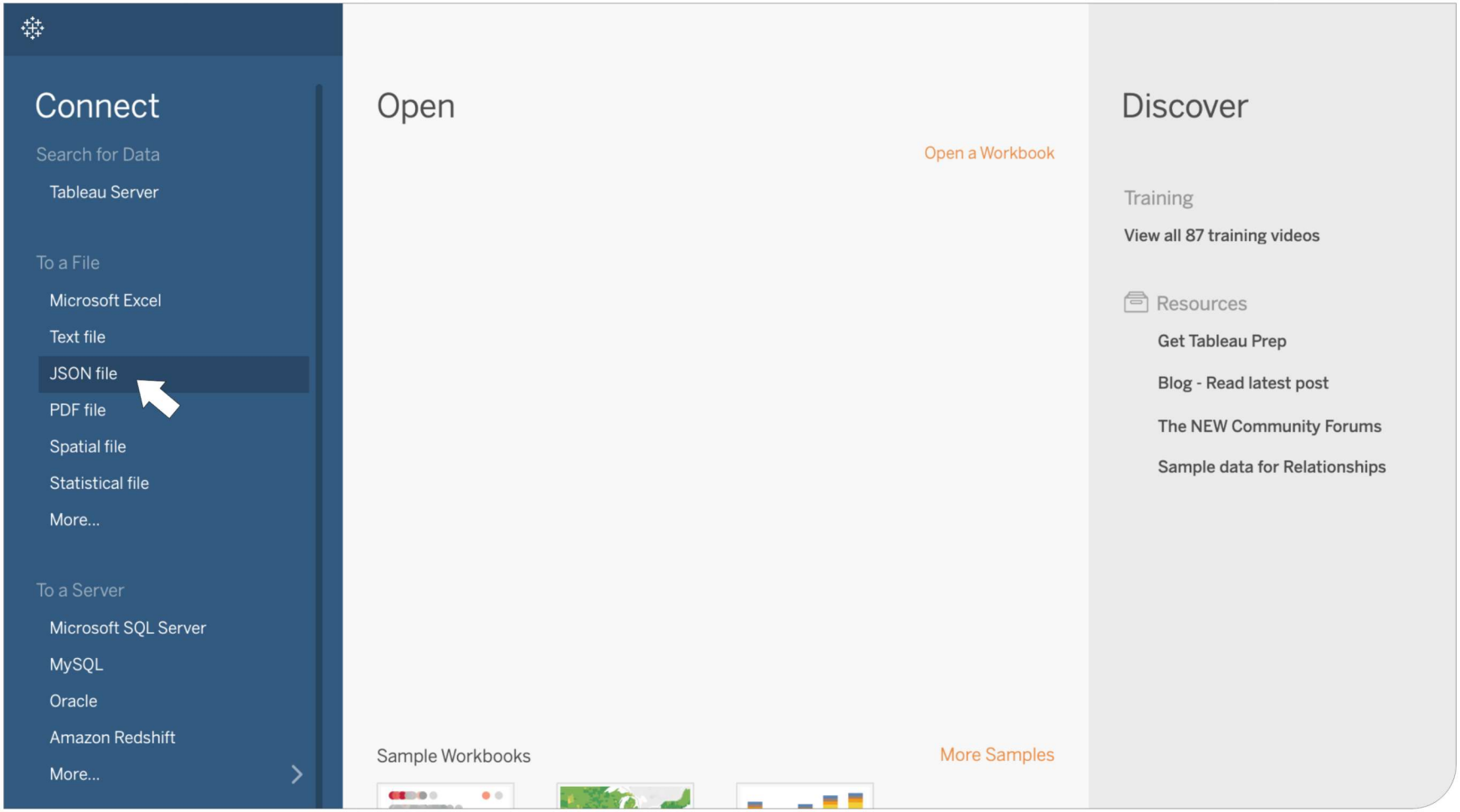
Import the data into Tableau

After installing Tableau and creating a trial account, your first step would be to import Daikibo's telemetry data.
You do that by first selecting a JSON file connection on the home screen of Tableau and choosing the JSON file with Daikibo's telemetry data.



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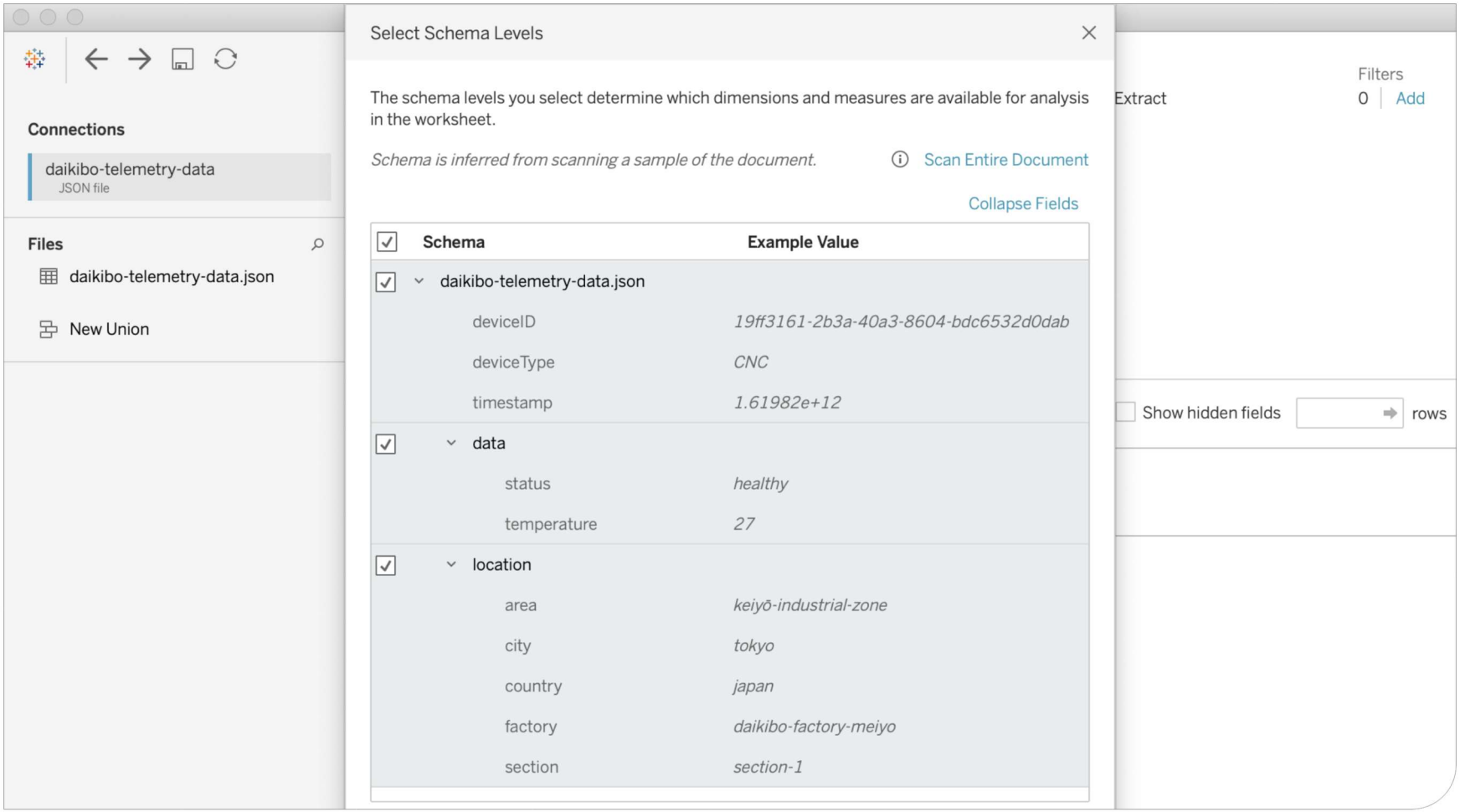
Import the data into Tableau

Then you need to check all Schema Levels as in the screenshot bellow. Press OK.



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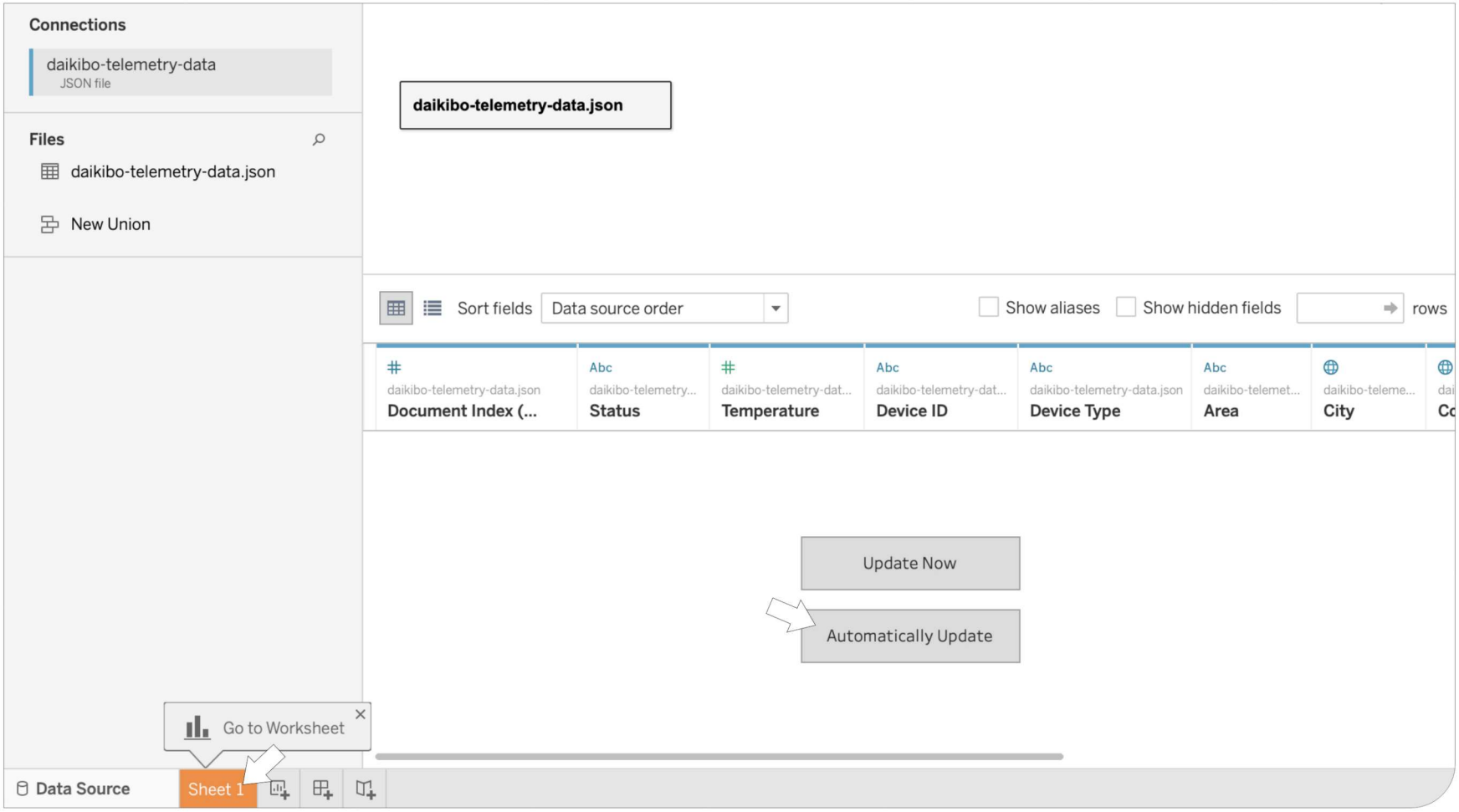
Import the data into Tableau

Press the Automatically Update button. The first 1000 rows will be loaded in the table.
Then open Sheet 1 (bottom left).



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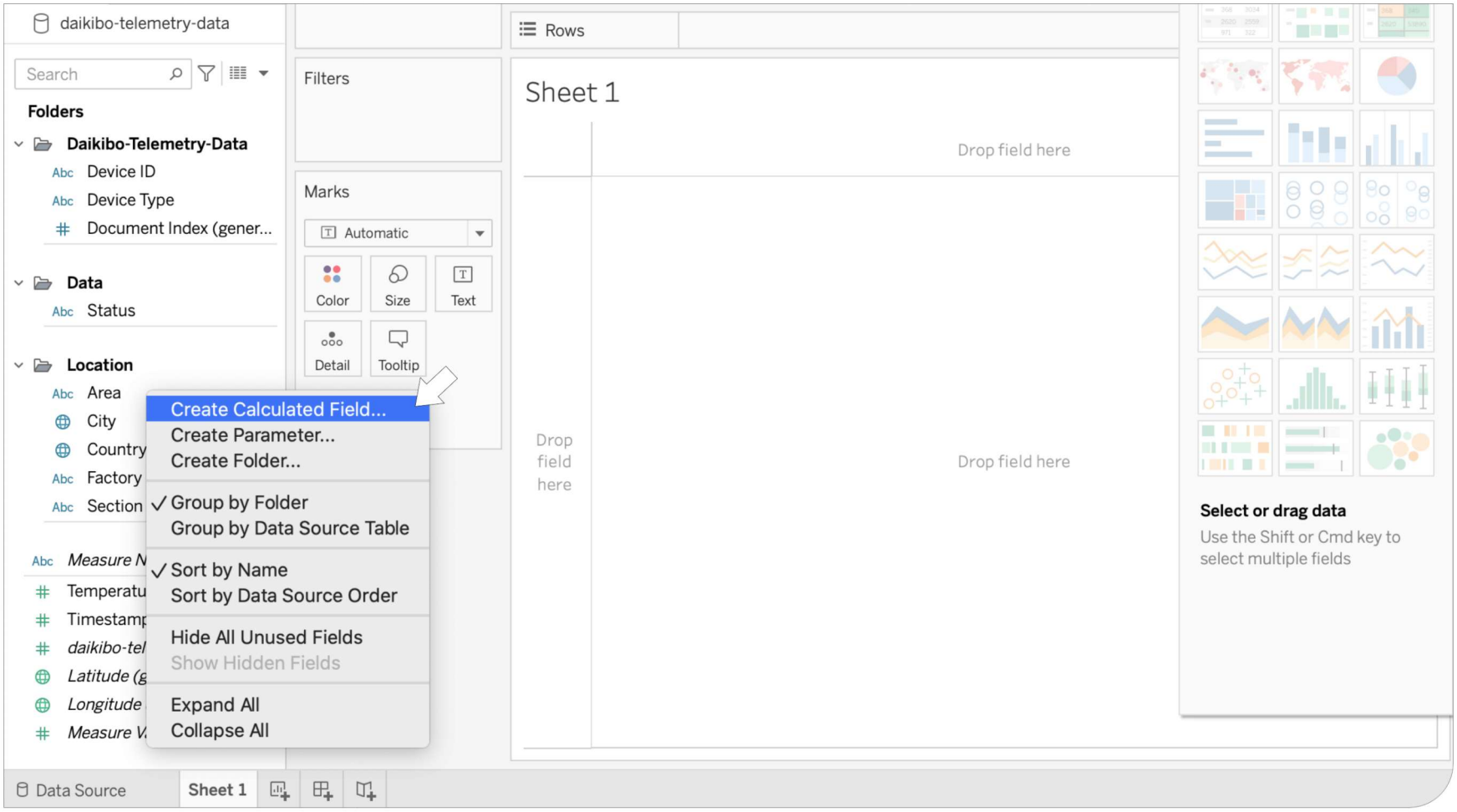
Create a Calculated Field

Under the last item of the Measure Names list - right click -> Create Calculated Field…



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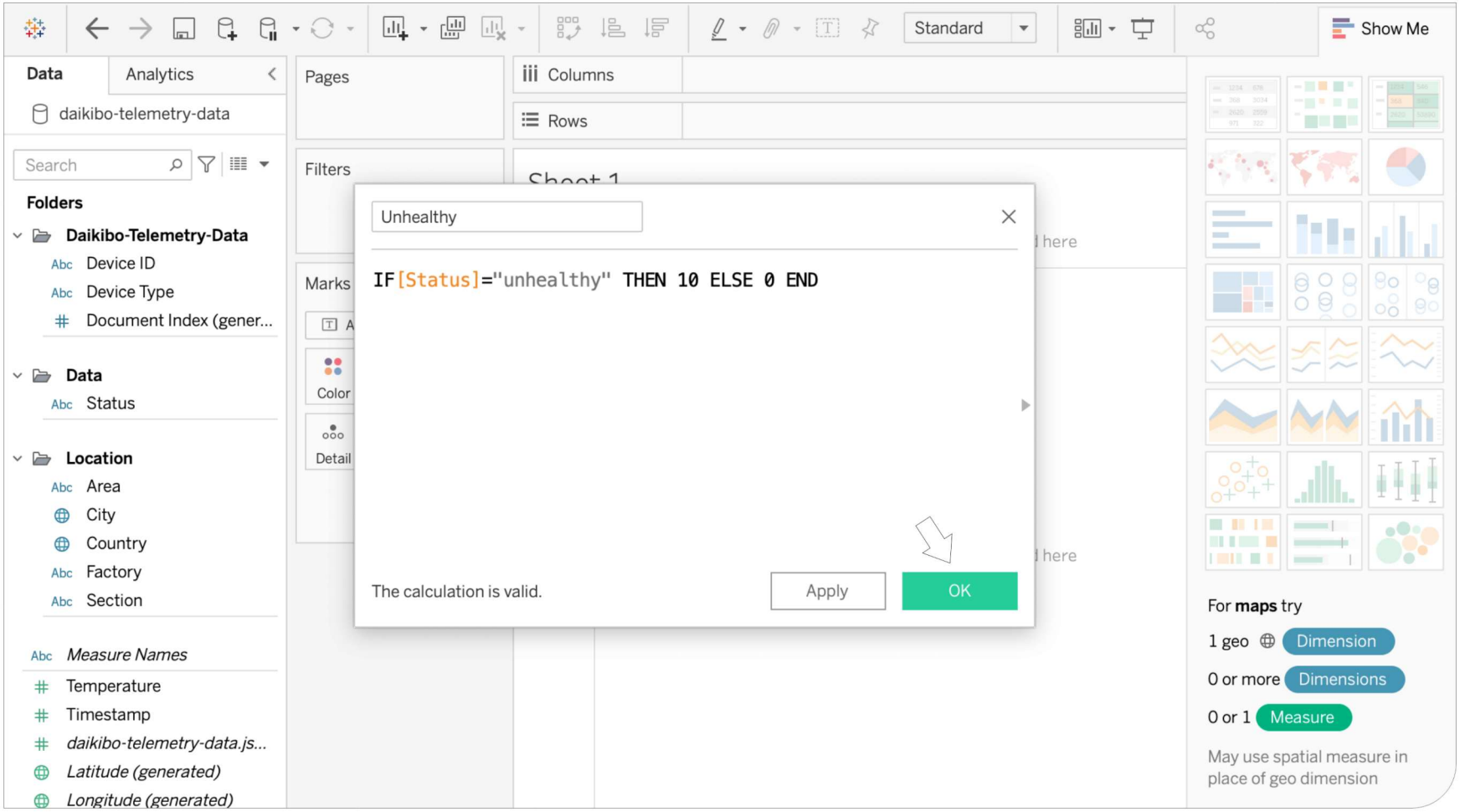
Create a Calculated Field

Name the field Unhealthy and fill in the formula bellow. Press OK to save it.
This calculated measure will help us learn how long was the down time of different pieces of the manufacturing process. Since messages are sent every 10 minutes, we set this measure to 10 for every unhealthy status received.



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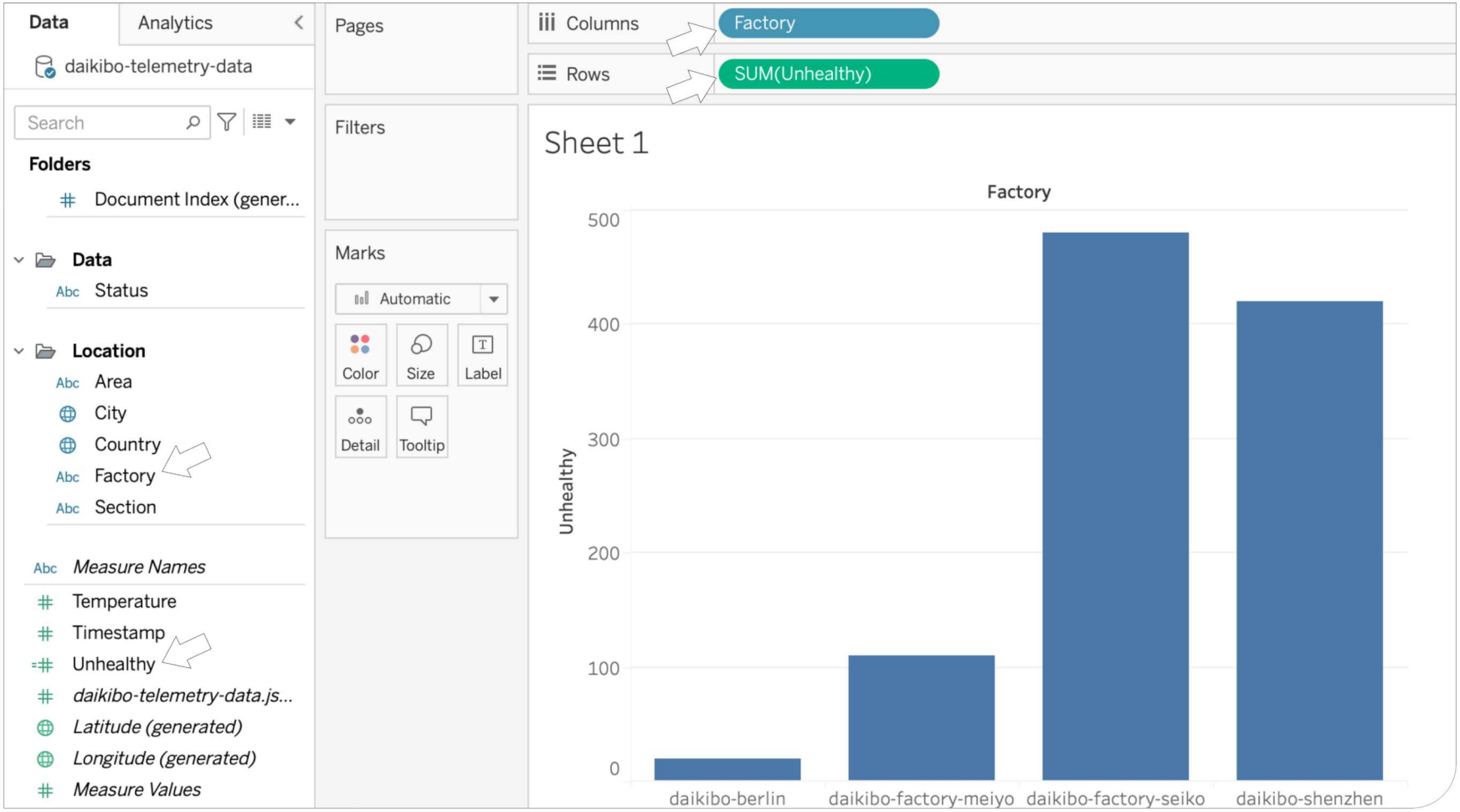
Create a Chart: Downtime Per Factory

The first chart we are going to create is Downtime Per Factory. We will use a classic bar-chart for that. First drag the Factory field under location to the Columns field (top middle). Then drag the Unhealthy field to the Rows field. If the factory names are not fully visible - just resize the chart by hovering at right edge and when the resize cursor shows up, drag it further right.



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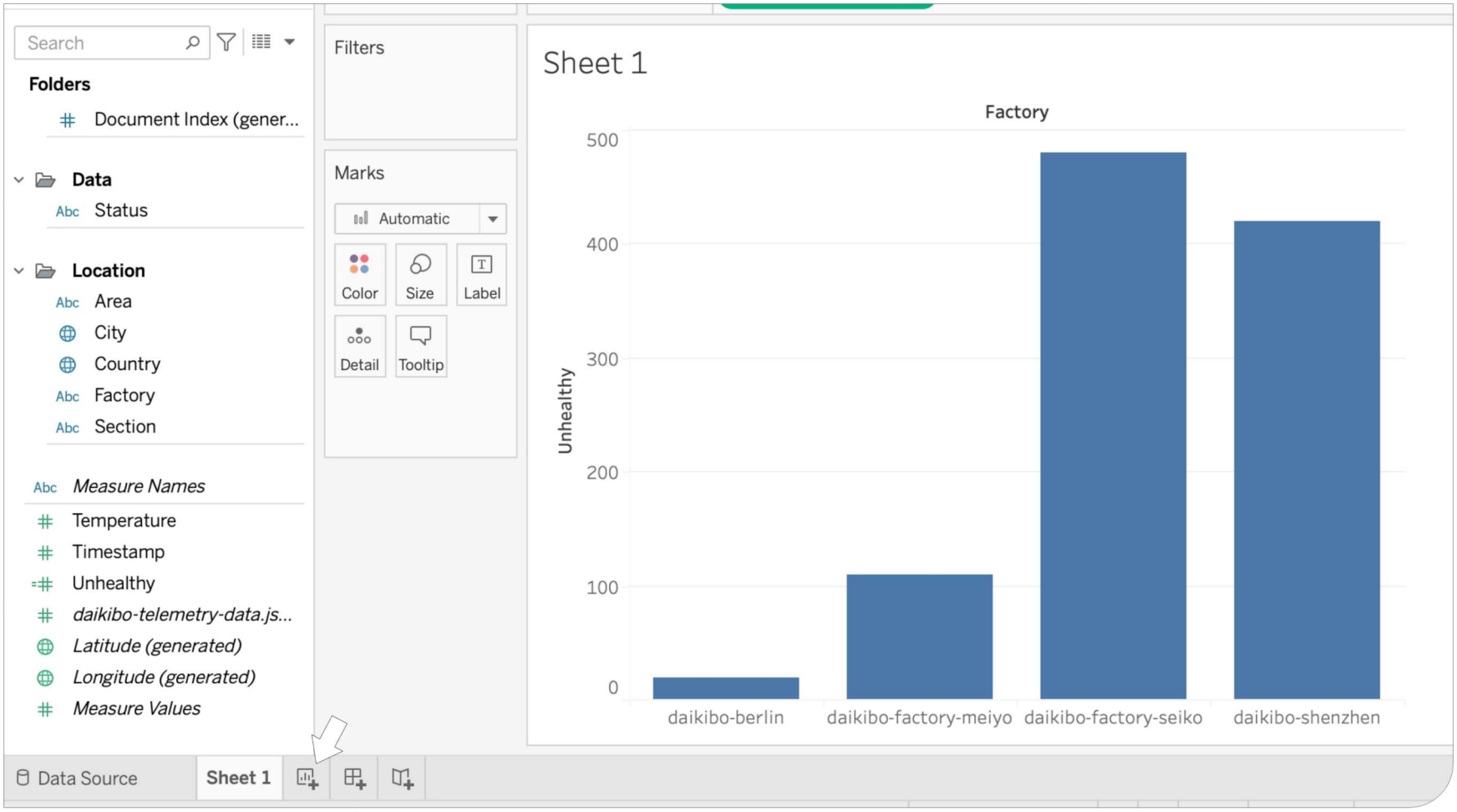
Create a Chart: Downtime Per Machine

Create a new sheet by pressing on the New Sheet button next to Sheet 1's tab (bottom left).



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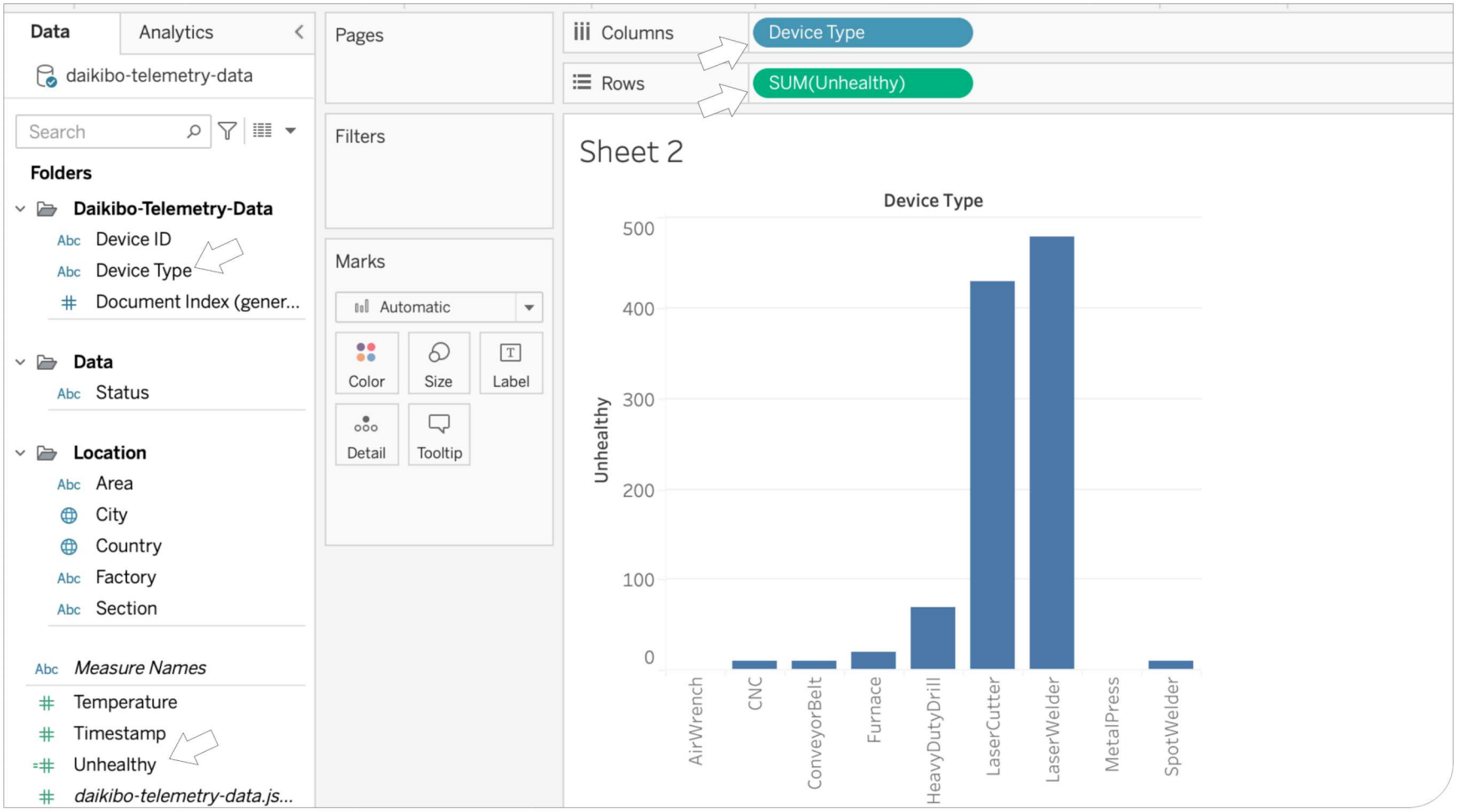
Create a Chart: Downtime Per Machine

In the new Worksheet, drag the Unhealthy field in the Rows field AND drag the Device Type field in the columns field.



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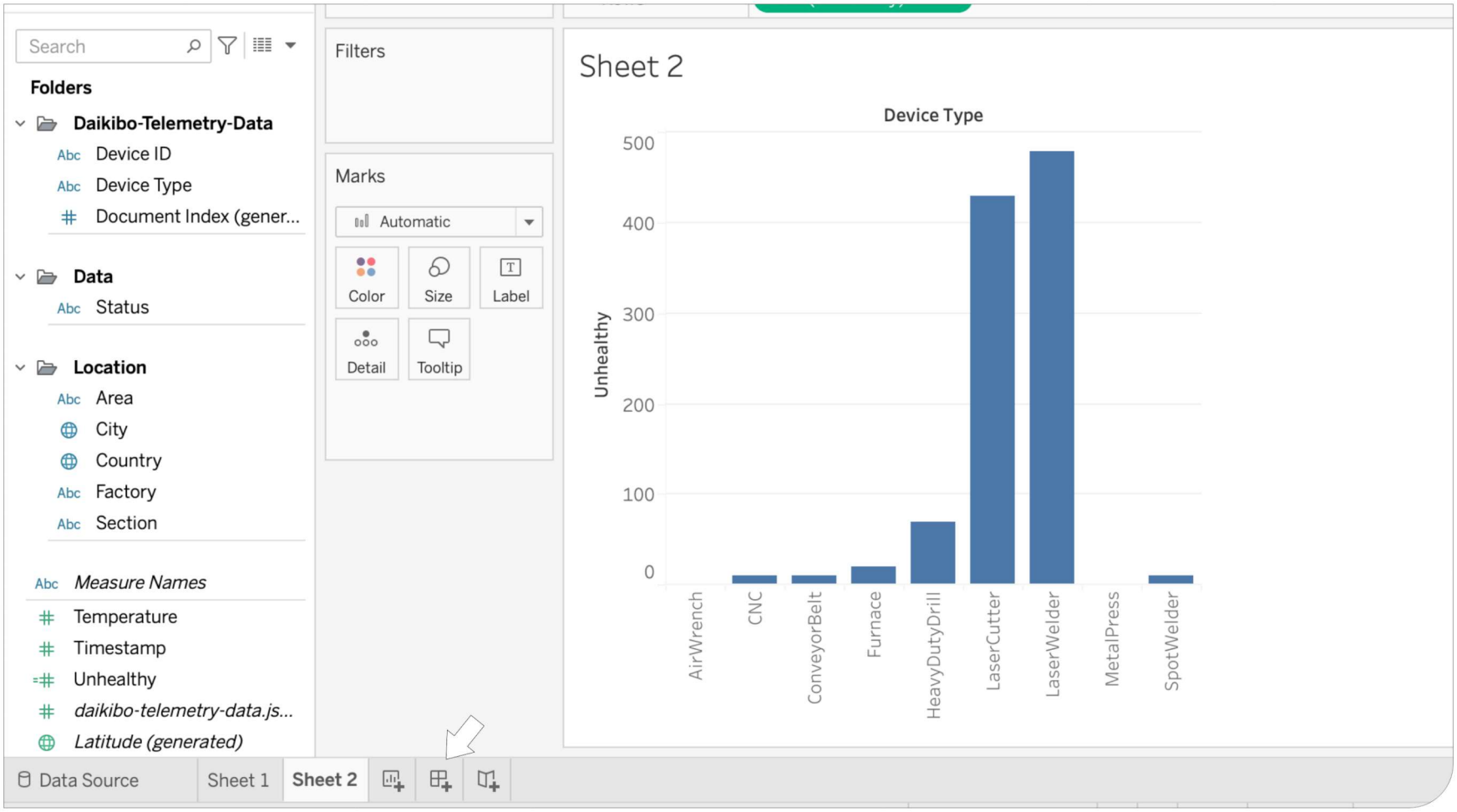
Create a Dashboard

It's time to combine those 2 charts to better piece together and understand the downtime at the different locations. For that purpose we will create a dashboard. Click on the button New Dashboard (on the right of the New Worksheet button we just used).



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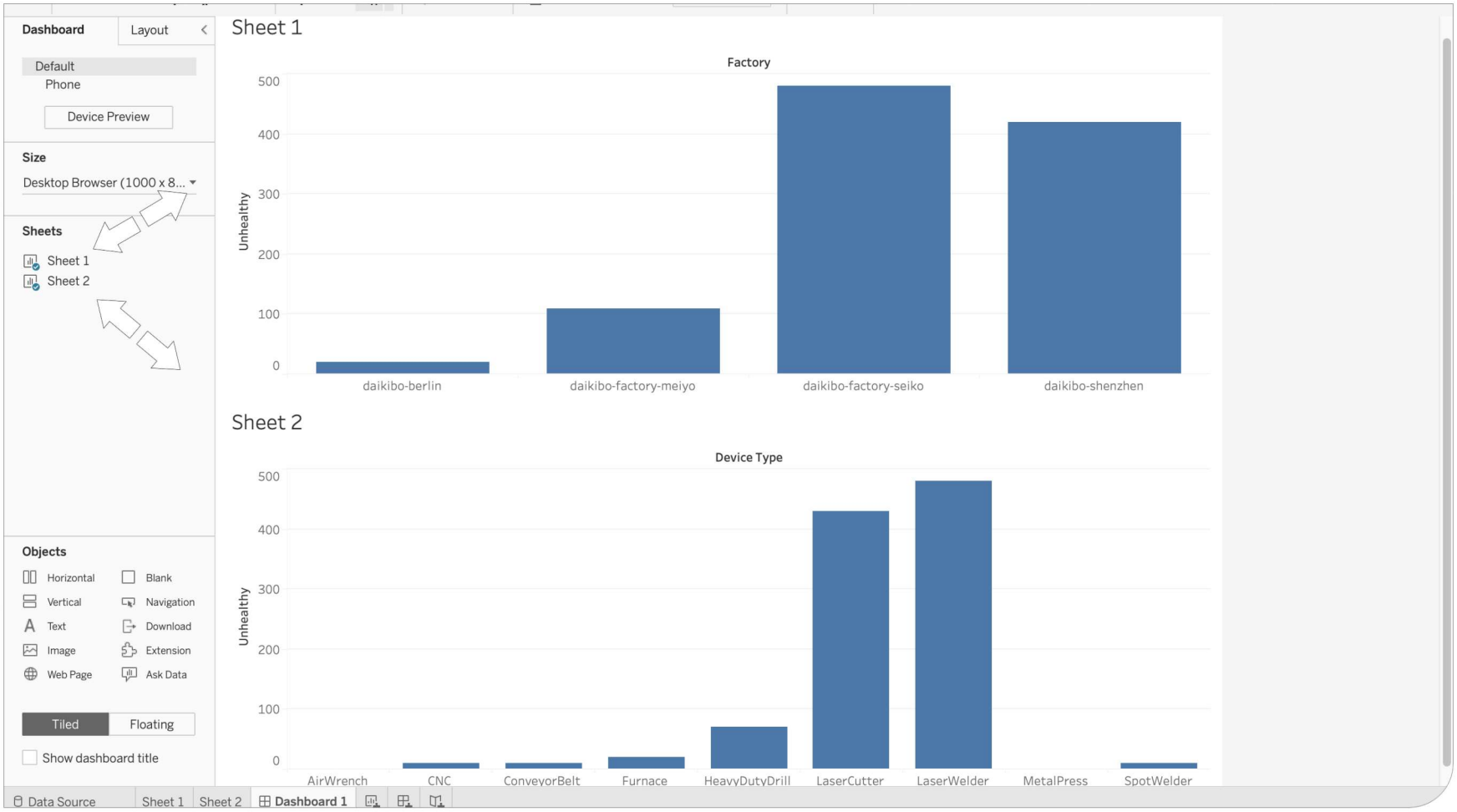
Create a Dashboard

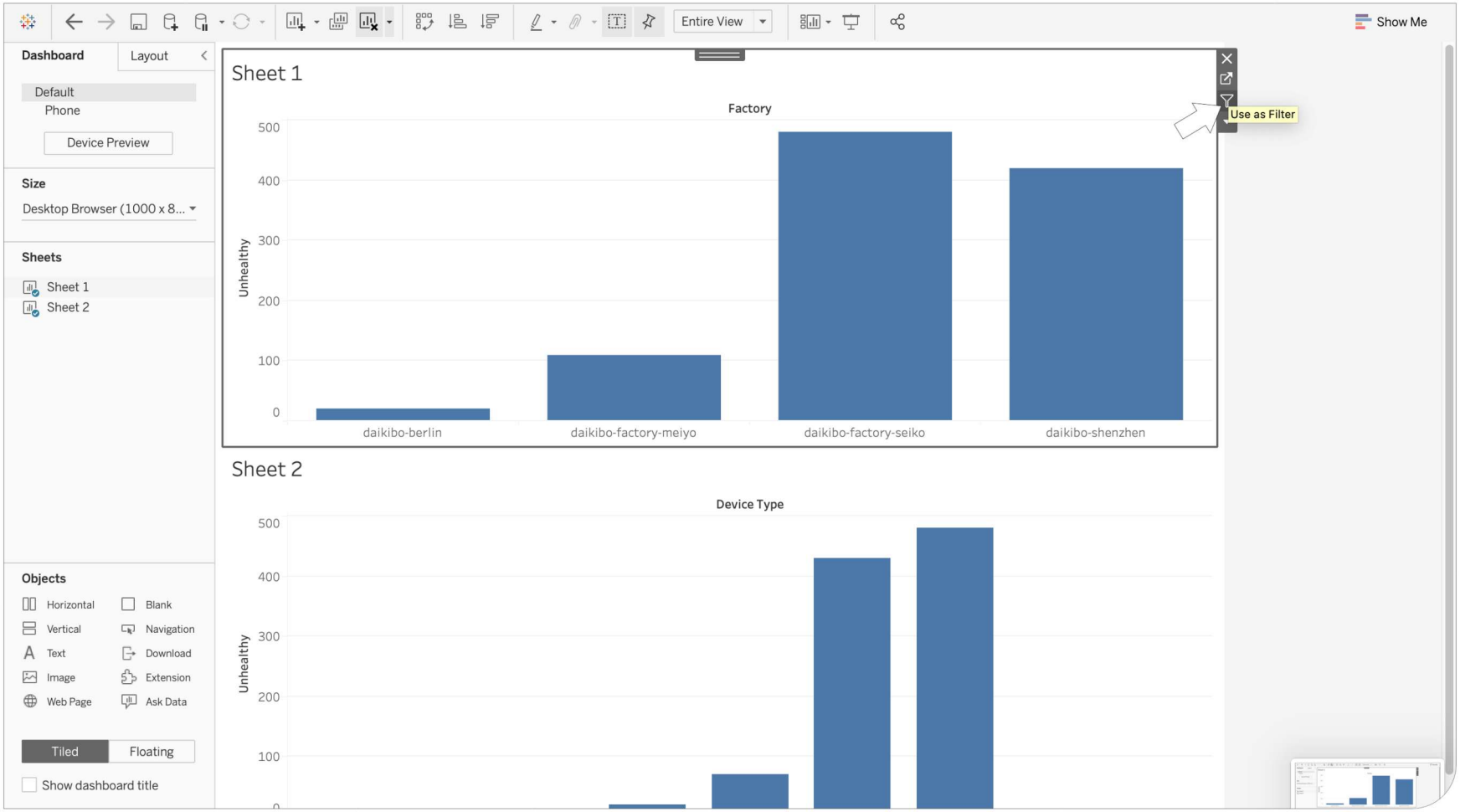
Drag Sheet 1 on top and Sheet 2 on the bottom of the dashboard's canvas.



Deloitte.

Create a Dashboard

And finally we want to sync the 2 charts and whenever we select a factory in the first (top) chart - the bottom chart to show only the down time of all machines there. To do that select the first chart by clicking once on it -> click the filter/funnel icon at the top right corner.



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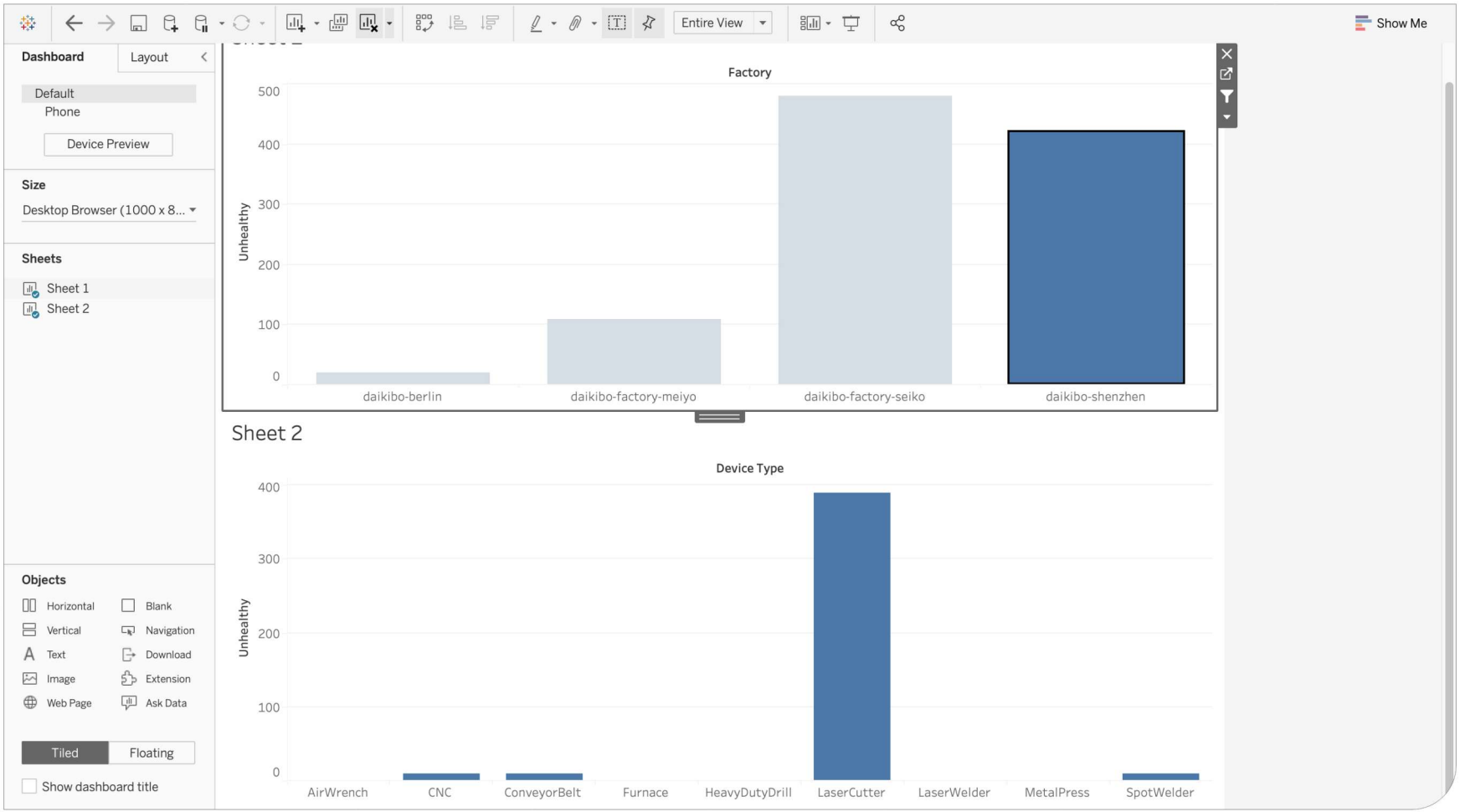
Create a Dashboard

Now when we select a factory, we can see what the downtime of the machines there, specifically, was. Here is what you should see if you select the Shenzhen factory.



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